1 Development Suite QlikView

1.1 General

This document is not considered to be a substitute for the QlikView help files but rather an addition. Specific elements are described and it is explained how to use them in MES-Cockpit. For any other information please refer to the QlikView help files by clicking the menu item "help" or "F1" in the QlikView editor.

The properties of the single elements can be opened and edited at any time by right clicking.

1.1.1 Data storage in Cockpit

Data presented in MES-Cockpit are divided as follows:

- Basic KPIs
  Basic KPIs represent the basis for calculating KPIs and also for displaying information on the different objects. This information is exported from all connected systems and saved as xml file in the directory \\<server>\QlikTech\Documents\keyfiguredata\ of the MES-Cockpit server. Basic KPIs include status information on objects as well as information recorded by HYDRA, e.g. posted quantities.
  Please note: Basic KPIs are used in the Performance Analysis and updated every 30 minutes by the MDC (Machine Data Collector).
- Master data
  Master data of the single objects identifying the different objects and also providing information such as defined authorizations and users. This information derives to some extent from the connected HYDRA systems but also from the MES-Cockpit administration system.
  The exported information is filed as xml file under the following path:
  \\<server>\QlikTech\Documents\masterdata\
- Online data
  Online data is used for displaying status information in Production Monitoring and include, e.g. the current status of workstations, the current ranking list of downtimes or the logged in operations. This information is exported from all connected systems and saved as xml file in the directory \\<server>\QlikTech\Documents\onlinedata\ of the MES-Cockpit server.
- Customer-specific storage
  All customized elements, such as customer-specific translations are saved as xml file in the directory \\<server>\QlikTech\Documents\custom\ of the MES-Cockpit server.

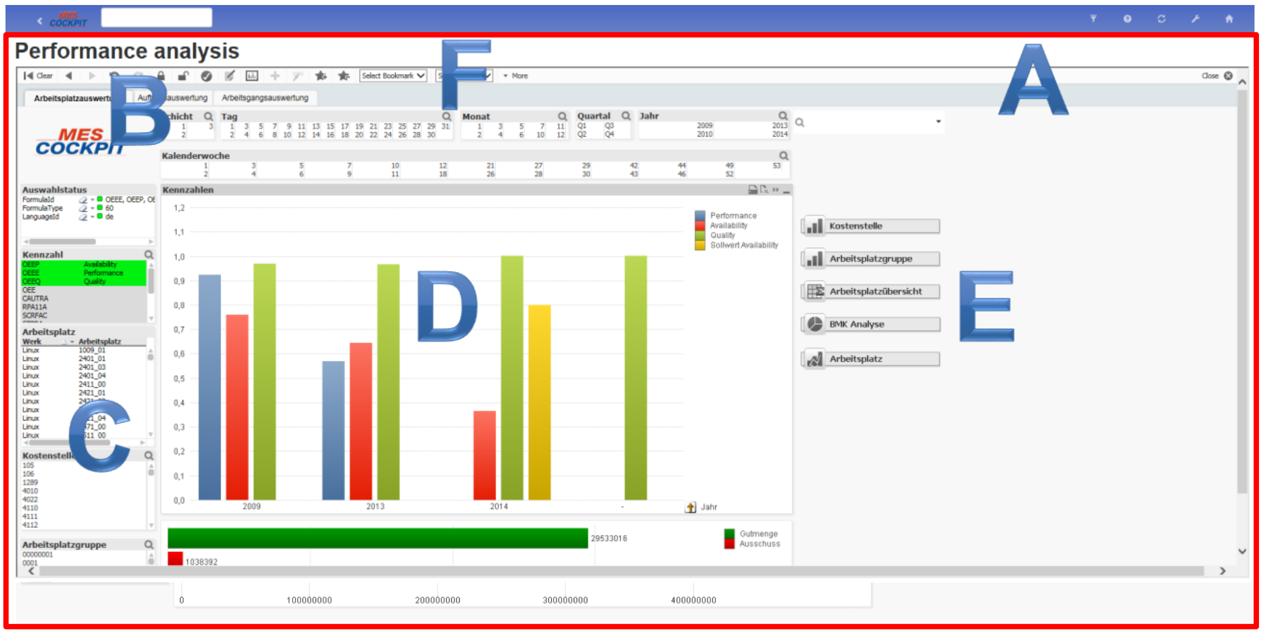
1.1.2 QlikView

A QlikView document allows for data from different sources and different formats to be used. This may be:

- Text files with separators
- Results of a database query, e.g. SQL via OLE DB and/or ODBC
- Other QlikView files
- QVD files (several files per QlikView file possible)
- Excel tables in BIFF format (Excel standard format)
- Files with fixed record lengths
- HTML tables
- XML files

1.1.3 Basic QlikView Elements

All elements of a QlikView file (*.qvw) are not only integrated but real parts of the file. This refers, for example, to: data, layout, worksheets, objects, etc. The layout of a QlikView file is, among other things, characterized by the following elements:



- A - QlikView document
  Each button/icon on the home screen represents a QlikView document that can be opened. They provide the data and worksheets including corresponding elements for evaluation.
  Further information can be found [here](#).

- B - worksheet tabs and thus the single worksheets
  A QlikView document uses at least one worksheet displaying objects. Even though a number of worksheet objects is located on different worksheets, they are still connected via the QlikView logic.
- C - Objects, such as list boxes / status boxes
  Objects included in a document can restrict data and/or the current restriction can be displayed. Restriction and, as a result, selection of data affects the entire document. This means, if machine 4711 is selected on the first tab, this restriction will be used for all elements where the machine is part of the data model.
- D - Diagrams
  Diagrams are the graphic presentation of selected KPIs. There are different types of diagrams.
- E - Minimized diagrams
  Minimized diagrams are diagrams that can be maximized by double clicking.
- F - Reports
  Using reports, selected data can be printed out in a defined format.

All other elements and relevant help files can be opened by clicking F1 if a QlikView document is opened or via the "Help" button.

1.2 QlikView Document

Each button/icon on the home screen represents a QlikView document or an application from the Production Information area.

Currently, the following QlikView documents are integrated by default in MES-Cockpit 3.1:

- Main document for the Performance Analysis:
  `\\<server>\QlikTech\Documents\MESC_Main.qvw`
- Main document for PerformanceMonitoring:
  `\\<server>\QlikTech\Documents\MESC_Online.qvw`

The original document must not be changed. Provided that modifications are required, the file has to be copied and renamed. Example: MES_Online_<abbreviation for customer>.qvw

Only then changes to this customer-specific file can be made.

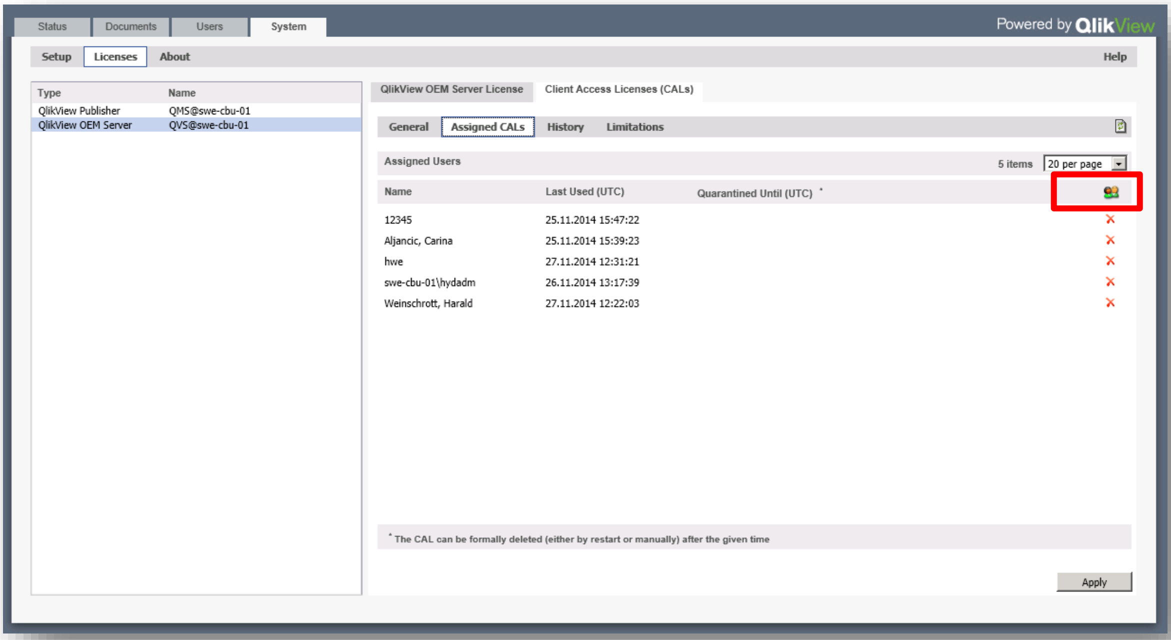
1.3 Start Processing

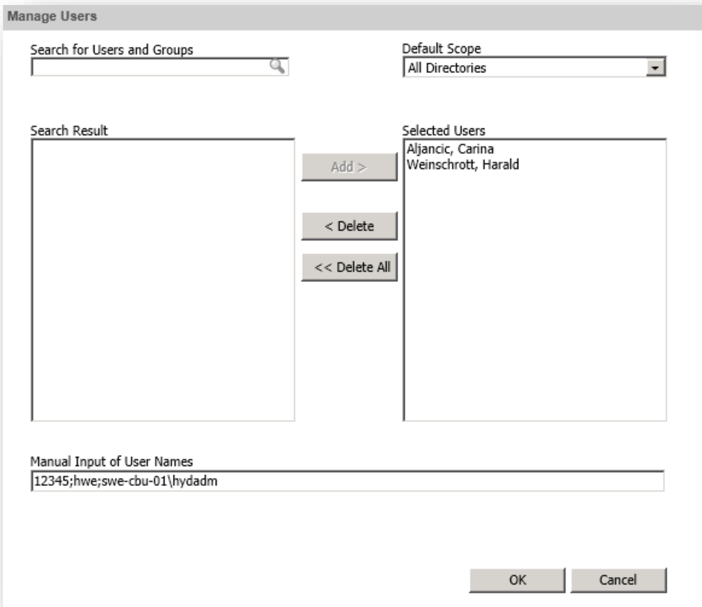
Processing can only be performed with a named license by QlikView. This license must be assigned to the corresponding user via the System Management Console. Only then a qvw document can be opened and edited by the user.

The current Windows login is used as the user's login data, i.e. the user will be verified.

Assignment of licenses

Assignments take place in the Management Console of QlikView in the MES-Cockpit server. The assigned users are shown and by clicking the user icon it is possible to remove and/or assign users





The license expires if a user does not "connect" to the sever for a long time. In this case, a qvw document must be opened in the server via the local client in order to connect.

Access is protected for standard documents. Consequently, they can only be opened and edited by users defined in the document. By default, the users hydadm and mescadm locally installed in the MES-Cockpit server are defined in the hidden script. Provided that the Designer is installed in the server, the document can be opened by the Windows login hydadm and the users allowed to edit the document may be defined.

The users hydadm and mescadm included by default should not be removed from the document in order to guarantee the default access.

1.3.1 Preparations

Step 1: Copying the required directories

qvw files should always be edited locally. Provided that data should be reloaded, the relevant *.qvw file and the corresponding sub-folders should be copied locally. It is recommended to copy the complete folder structure of the Qliktech folder from the server (C:\ProgramData\QlikTech) into a local working directory

Please note: This folder also includes the exported xml files. Single sites or time frames can be omitted if not all data is required for the design.

Step 2: Authorizations and responsibility areas

This step can be skipped if the user is already set up in MES-Cockpit and has relevant function authorizations.

After copying, the user of the *.qvw file (Windows user) must be assigned to the specific customer, provided it has not yet been defined as user in MES-Cockpit. To do so, proceed as follows:

- The following authorizations have to be defined for the user in the local file Default.xml (…\MES-Cockpit\custom\responsibility\Default.xml):
  - MC-WPAnalysis
  - MC-OAnalysis
  - MC-OPAnalysis
  - MC-Overview
- Then the corresponding buffer.qvd file (…\MES-Cockpit\custom\responsibility\buffer.qvd) has to be deleted. This file is generated once more by the following action:

- Start the MESC_Loader.qvw and reload the file using the shortcut CTRL+R. Afterwards the file can be deleted once more without saving.
  The sole task of the file is to generate the required qvd files.
- Then the qvw file to be edited can be opened.

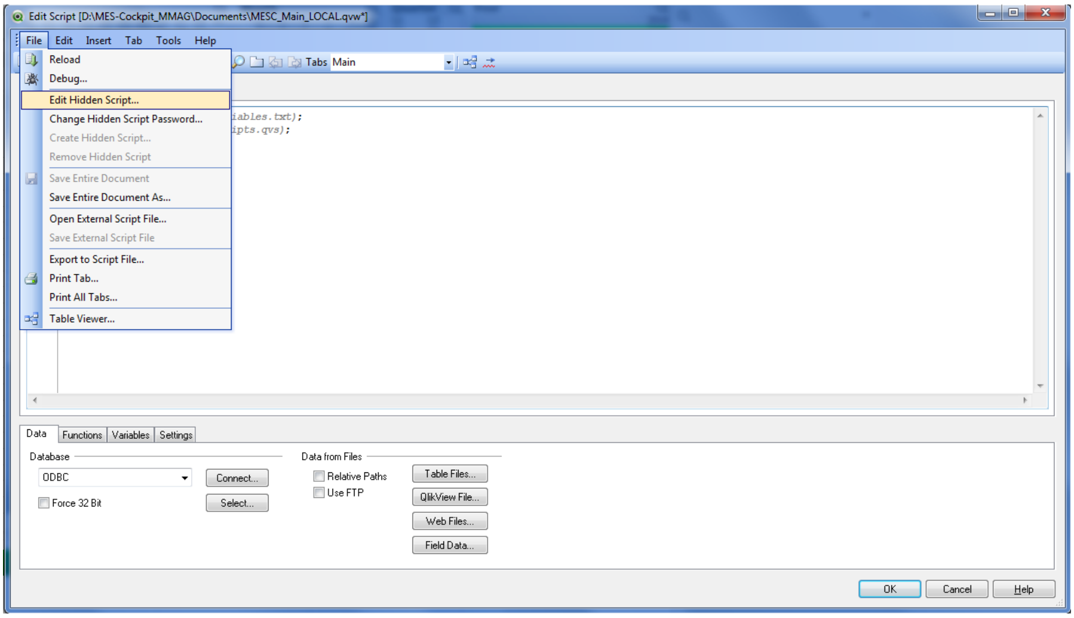
Step 3: Defining authorizations in the hidden script

This step must only be carried out in server documents. Authorizations have been removed from the qvw documents of the CUT package in order to avoid this step.

In order to enable further users to edit qvw documents, they must also be defined in the hidden script. To do so, proceed as follows in the QlikView document:

Open the menu item "edit script" via File --> Edit Script (CTRL+E) and then edit the hidden script via File --> Edit hidden script.

The password opening the hidden script is: `Mosbach 74821`



The hidden script includes as of row 6 the users allowed to edit the document. If an additional user is required, the new entry must comply with the structure of the existing entries.

```
<Domain name>\<User>, ADMIN, *, *
```

Example:

```
MPDV\USERNAME, ADMIN, *, *
```

After saving, the file has to be reloaded using the shortcut CTRL+R. If all steps succeeded, the user can open the qvw file and view the default tabs including data after reloading (CTRL+R).

1.3.2 After Processing

After processing, the revised, customized qvw document can be restored in the server and is directly active and displayed.

Only the qvw document may be restored on the server, as all other data are refreshed in cyclic intervals and are therefore more up-to-date.

Apart from restoring the qvw file, the steps mentioned in section "Calling up a new evaluation/report with an existing button" from the document MESC_DevelopmentSuite.pdf are still to be carried out.

2 Designer

2.1 QlikView Document Properties

General

General properties of the document, e.g. title and author of the document.

Open

This tab enables different configurations in order to open the document. For example, it can be specified if a picture is shown while opening the file.

Worksheets

The "sheets" tab of the document properties helps you to keep track of the document's worksheets and objects. The dialog includes two lists. A list of the worksheets in the upper part and a list of objects in the lower part. By clicking, the lists can be sorted by any column.

Server

This dialog includes document properties affecting usage in a QlikView server.

Update

If the document is provided in a QlikView server, you can configure an automatic update by executing the script at regular intervals via this tab.

Variables

Listing of all variables used in the document and/or defined for the document.

Security

Here you can define the properties for the users' rights for documents. By default, all options are selected (enabled). Users with administrator rights may access this dialog at any time and change the settings. But the settings can prevent other users from changing the layout of the document.

Trigger

In the "trigger" dialog, you can define that activities (including macros) are executed with specific events (relating to the document or single fields or variables).

Groups

Please note: The tab "groups" is only available if the document includes data and the script has been reloaded once.

This tab allows you to divide fields of the document into groups.

Tables

When data that includes circular references is loaded into QlikView, loosely coupled tables are created automatically. This avoids that the circular references create a loop in the QlikView internal logic. These loosely coupled tables need to be handled in order to visualize data in a way that is expected and understandable.

Sort

The properties tab allows for the sort sequence of each field to be configured. (You can also define the sort sequence in the properties of the object: define sorting).

Presentation

Here you can define the default settings for presenting field values for created list boxes and multi boxes. Available fields are listed in the "fields" group.

Numbers

This properties tab provides number formats for all fields and variables of the document.

Encryption

Only users with admin rights can access this page. You can encode the data of one or several fields.

Modifications

In this tab you can select modifications to change the layout of the document.

Please note: In QlikView documents active modifications only become effective in the AJAX client or the WebView mode.

Font

You can set the font type, font style and font size in this dialog.

Layout

The page "layout" exists in the properties dialogs of objects and documents. Consequently, the settings refer to a single object or to all objects of the document.

Title bar

The page "title bar" exists in the properties dialogs of objects and documents. Consequently, the settings of the "properties" page of the document refer to a single object or to all objects of the document.

On the page "title bar" you can configure the title bar of one or several objects.

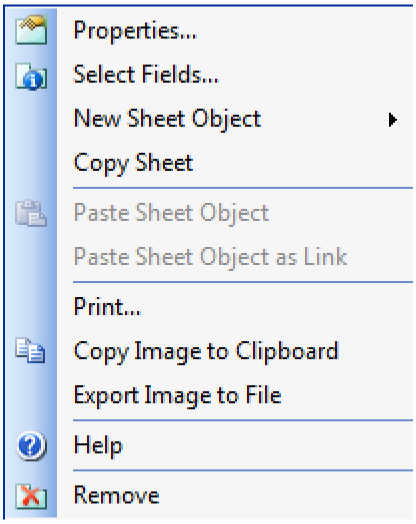
2.2 Worksheet

The worksheet menu is opened by right clicking an empty space on the worksheet or, if the worksheet is active, via the menu below the object on the menu bar.

A QlikView document uses at least one worksheet displaying objects. Even though a number of worksheet objects is located on different worksheets, they are still connected via the QlikView logic.

The menu provides, among others, the following items:

- Properties: leading to the properties dialog of the worksheet. Here you can select a background image or change the layout of objects on the worksheet.
- New Sheet Object: a new object can be inserted in the worksheet. To do so, a list of available elements that may be inserted opens.
- Copy Sheet: the currently selected worksheet is copied 1:1 and added as the last tab to the existing tabs.
- Remove: deletes the worksheet including all displayed objects



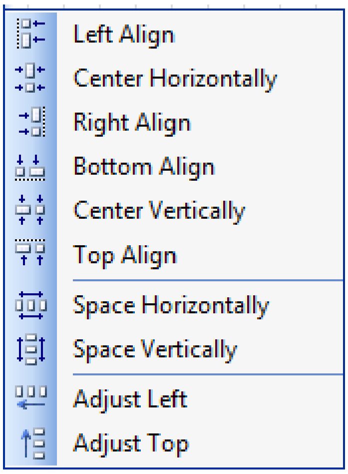
2.2.1 Worksheet Properties

The following worksheet properties should be emphasized:

- General
    - Show workplace - condition: an authorization necessary to view the workplace may be defined here.
      Example from the workplace analysis:
      `=$(ISAUTHORIZED('MC-WPAnalysis'))`
- Security: the user's properties/rights can be defined in the "security" tab. Thus, a user can be prevented, for example, from changing the size and position of displayed objects.

2.2.2 Formatting of the worksheet

Objects included in the document can be aligned using the menu item "Layout" --> "Align/Distribute"



Please note: Several objects can be selected at the same time using the mouse pointer and a window or by holding the shift key.

2.3 Objects in QlikView

QlikView objects are the elements used for the selection and display of selected data. There are different elements.

New objects can be inserted via the context menu in the worksheet "New object". If several inserted objects are selected they can be aligned via the context menu.

The following objects are available in QlikView and/or can be integrated:

- List box
- Table box
- Status box
- Text box
- Charts

The properties of single objects can be opened and edited by right clicking the objects on the title bar.

2.3.1 List Box

The list box is the simplest and most common object used in QlikView. It represents a field of a data source and includes a list of all field values. It has mainly been designed to select and restrict the displayed data. All displayed values are presented "distinctly"

List box properties

Can be started by right clicking the header.

- General: General properties, such as the title of the list box

- Formulas: Formulas presented in the list box, whereas each formula corresponds to a column in the list box
- Sorting: Definition of the sort sequence of values
- Presentation: Configuration of the list box displayed in the worksheet
- Numbers: Definition of number formats specific to the list box
- Font: Definitions for presentation
- Layout: Definition
- Title bar: definitions for the title bar and available functions in the title bar

Inserting a new list box

Right mouse click the worksheet --> New object --> List box

Example: *Creating a new list box including machine descriptions and the following characteristics:*

- *Definition of a multilingual title*
- *The title bar should show the search icon and the Excel export icon*

Solution: Choose wm.designation (machine description) in the drop-down list of the field and define the following in the "title" field: =$(LOCALIZE('lkWorkplaceDesignation')). You also have to select the "search" icon and the "Export to Excel" icon in the title bar tab.

2.3.2 Table box

The table box is an object showing several fields at the same time. Values are displayed in data records, i.e. related values are displayed in one row. Any fields can be combined in a table box, even though they derive from different data sources.

Table box properties

The properties of a table box can be opened by right clicking the header (context menu). The following options are provided:

- General
  - Title: title of the element "table box" on the worksheet
  - Available fields: fields of the table box
    - List of fields provided by the data model
    - Inserted fields displayed in the table box
  - Settings for the selected field
    - Name: definition of the field name for the field selected in the list of inserted fields. The function LOCALIZE allows for a LanguageKey to be defined
- Sorting: definition of the sort sequence of displayed information

- Presentation: definitions for display and presentation of information, e.g. left-aligned and if a drop-down box should be used
- Design: definitions for the display, e.g. the color of the table box
- Numbers: definitions for the display of numeric values, e.g. two decimal places
- Font type
- Layout of the complete object, e.g. with shadow effect
- Title bar: definitions for the title bar and available functions in the title bar

2.3.3 Status box

The status box lists statuses by the name and value of fields. This tool shows the same information as the "selection status" dialog. But like any other object, it is directly located in the worksheet. The "indicators" dialog is used in order to differentiate between selected and blocked values. In MES-Cockpit the status box has been integrated with the title "selection status"

Status box properties

The properties of a status box can be opened by right clicking the header (context menu). The following options are provided:

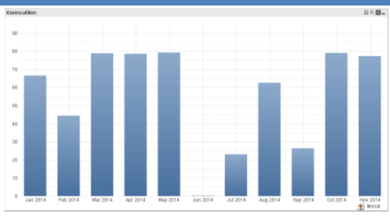
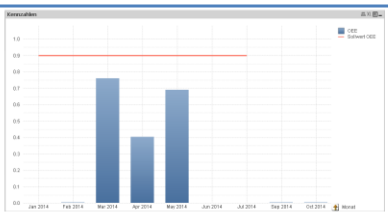
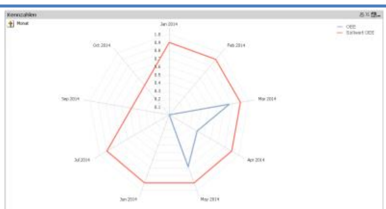
- General
  - Title
    Caption displayed in the status box. A multilingual text can be defined via the LOCALIZE function. Default example:
    `=$(LOCALIZE('lkSelectionStatus'))`
  - Displayed columns
    The columns displayed in the status box can be selected.
- Layout
  - Shadow
    Defines how the status box is presented on the worksheet.
- Title bar
  - "Show title" checkbox
    If this option is enabled, a title bar including defined title is shown.
  - Title text
    The defined text corresponds to the title of the "general" tab.
  - Icons
    The parameters displayed in the status box can be selected.

2.3.4 Text box

Text boxes have been designed to display any texts or pictures on the worksheet in order to improve the layout of the document. They can be inserted at any position on the worksheet even in areas including other objects.

2.3.5 Charts

Charts are graphic presentations of numeric data. The chart type is defined in the dialogs "chart properties", "general". The following chart types are available:

Chart type	Description	Sketch/example
Bar charts	Particularly suitable for comparing similar values. Common and plain chart type.	
Line chart	Line charts present data as lines between the single values, either only as single values or as lines and values. This chart type is especially useful to present developments and trends.	
Combination charts	The combination chart is a combination of bar charts and line charts. The values of one formula can be presented as bar chart and the values of another formula may be displayed as line or scatter chart.	
Radar charts	A radar chart is a line chart with the x-axis aligned on a circle (360°). Separate y-axes are radially aligned for each variable of the x-axis. Sometimes this chart type is also referred to as spider chart.	

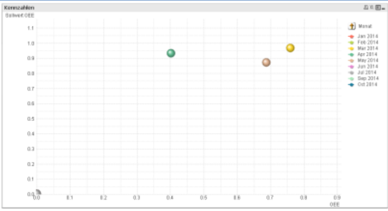
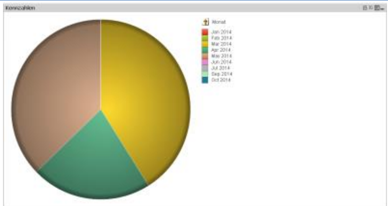
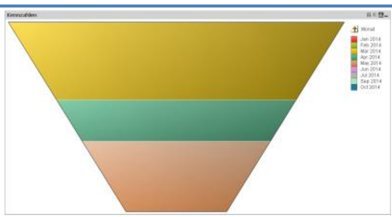
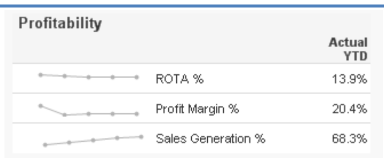
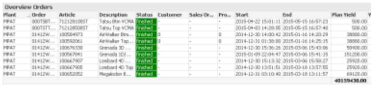
Gauge chart	Gauge charts show the result of a single formula without referring to the field values.	
Scatter charts	The scatter chart presents value pairs from two formulas. This is useful in cases when two formula values should be assigned to each dimension value (e.g. population and growth of population for each country).	
Pie chart	Pie charts usually represent the relation between a single dimension and a formula. But you may also define a second dimension.	
Funnel charts	The funnel chart is especially useful if distributio data and process data should be displayed. It is similar to the pie chart. The segment's height (and/or width for horizontal orientation) or surfac depend on the value of the formula. Alternatively the same height or width can be used for the segments. But it is recommended to show nume formula values.	
Mini charts	Small diagrams included in table cells enabling compact comparisons.	
Pivot table	Pivot tables are a powerful means of data analysis They have extensive functions but they are still easy to operate. Pivot tables show dimensions and formula values in lines and columns, e.g. as cross-tabulation. Data may be grouped in different ways. In addition, partial sums can be displayed in pivot tables	

Table chart	In contrast to pivot tables, table charts cannot display partial sums and they cannot be used as cross-tabulation. But each column can be sorted as required and each row includes a combination of dimension(s) and formula(s).	

2.4 QlikView reports

Data can be printed out via the print function or a report. A report summarizes several charts and tables to display and print them together.

QlikView provides a Report Editor allowing to combine different objects of one or several dialog pages into a clearly structured report and to define the corresponding headers and footers.

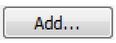
QlikView differentiates between four types of reports:

- Document
  They are created and saved within a QV document. Users can access the documents locally or via the QV server.
- Users
  MES-Cockpit does not allow creating user-related reports.
- Personal
  MES-Cockpit does not allow creating personal reports.
- Shared server
  MES-Cockpit does not allow creating reports on a shared server.

In the local client reports can be managed in the "Reports" menu of the Report Editor. A list of available reports is shown and new reports can be created by choosing the option "Edit reports". MES-Cockpit, i.e. the web client, does not allow creating new reports but existing reports can be opened.

2.4.1 Designing reports

The Report Editor opens and the following options are available when clicking the menu item "Reports" --> "Edit Reports":

- Create new report (,  or via the menu item "Reports" – "Insert")
- Delete existing reports
- Change report settings

Report settings

Starting via "Reports", "Report settings" including the following contents

- Settings: settings affecting the complete report, such as title, paper size and viewing conditions
- Margins: definition of margins in cm in the report
- Header/footer: configuration of headers and footers for the entire report
- Selection: defining the data to be displayed in the report:
    - Current selection: selected data is displayed in the report and can be printed
    - Unselect: all selections made are canceled and data is printed without selection
    - Bookmark: the bookmark selected in the drop-down list is started before printing. Original selection will be restored after printing

Once a report has been created, it can be switched to the Data Sheet Editor by double clicking the report list or via the button "Edit >>".

Data sheet editor

The Data Sheet Editor allows "designing" the selected report. Reports are designed by inserting data sheets. The following two types are differentiated:

- One-sided data sheets:
  They may include any number of objects. The one-sided data sheet is always printed on one page. The objects must be adjusted to the page size.
- Multi-page data sheets
  They only include one object that might extend over several pages. The number of pages depends on the number of data to be printed.